Profitable Work First. Compounding Capability When Worthwhile.

Transformational Intelligence and the Network That Invents Its Economy

Find your Alpha. Build your Successor Ω.

AGI ALPHA independent research lane

Prepared with AI assistance for scientific review.

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Abstract

We investigate whether competent frontier intelligence can profitably deliver complete proof-carrying software components at buyer-chosen prices referenced to equivalent human services. A finite-cash simulation separates paid delivery, additional capability retention and optional research. Buyers can choose a human service, wait or decline. The tested graph lifecycle guards scoped admission and network reuse without changing existing settlement authority. We execute 1,728 training histories and 4,560 fresh confirmation histories. Model/API prices and labor sources anchor costs; task capability, human effort, demand and skill effects remain explicit assumptions.

Primary finding. The best tested ordinary-service policy charged $10,000 and made no retained-capability investment. In the reference profile, 14 fresh histories produced a pooled full cost of $1,058 per useful outcome against a $23,714 cost-derived human reference; mean complete profit was $151,024 and available operating funds $122,040 over 72 business weeks plus runoff. All 14 were profitable. The same price produced no orders and cash closure in all 13 burdensome-profile histories. Optional retention reduced reference-condition mean profit by $65. Three network-admitted assets in the high-repeat/slow-Beta condition did not recover their direct retention costs. The supported positive result is conditional profitable Beta service; capability compounding and realized commercial success are not established.

Evidence boundary. The simulator, cryptographic fixture checks and lifecycle tests execute. The customer jobs, collections, Alpha qualifications, transfers and investments are modeled. No actual customer revenue, frontier-model production calls, live ENS authority or production-scale operation is claimed.

1 Question, mechanism and independently testable claims

The primary question is whether an automated network can sustainably earn complete-cost operating profit and available surplus by delivering demanding, useful accepted work. We answer it before asking whether retained mission intelligence or further investment improves the enterprise. Profitable Beta-only operation is a positive commercial result; it needs no graph advantage or prescribed investment count. We distinguish two sources of advantage. **Beta** is competent use of current frontier models and ordinary tools, including decomposition, adequate context, source grounding, implementation, proof software, critique, correction and recovery. **Alpha** is the additional net mission advantage of a retained capability over that Beta and the useful capabilities already available to a recipient.

The commercial comparison asks whether complete accepted Beta delivery is economical relative to equivalent human service. The incremental comparison asks whether retaining a method improves the same enterprise. An attractive human–AI price gap cannot establish Alpha. Conversely, a failed retention project does not invalidate profitable ordinary service. Our experiment permits both outcomes and does not require a fixed investment sequence.

The primary pathway is paid useful work, possible learning during delivery, optional incremental extraction and qualification, admission if net Alpha is supported, optional reuse by genuine beneficiaries, then fresh work under the current frontier. The customer buys an outcome, not an experiment designed solely to generate graph nodes. A demanding job may leave no reusable method; an apparently reusable method may be redundant, fail fresh verification or become obsolete before repayment.

Claim	Required evidence in this study
Useful automated delivery	Useful accepted outcomes under the declared supported scope; failures and false passes retained.
Beta service economics	Complete cost and cash viability at buyer-chosen prices, with a transparent matched human-service reference.
Retained Alpha	Fresh paired net advantage over same-version Beta and relevant inventory after additional retention/reuse costs.
Network value	Shared availability improves enterprise outcomes relative to creator-local availability at equal agents and resource budgets.
Moving-frontier persistence	Fresh qualification after model/tool changes supports remaining value before expiry, or the skill retires.
Productive feedback	Paid activity supplies resources or capabilities that improve later work after their complete incremental costs.
Self-acceleration	Separate evidence that later worthwhile invention or validation becomes more effective at comparable difficulty; job-volume growth is insufficient.

Table 1: Separate claims prevent one favorable number from becoming a general “Flywheel success” label.

2 A demanding venture chosen before outcomes

We independently screened four complete R&D offerings: proof-carrying bespoke software, simulation-native inverse design, hardware-aware algorithm discovery and coordination-mechanism design. The prospective criteria were useful executable value (25%), autonomous falsifiability (25%), distinguishable incremental Alpha (25%), a clean moving-Beta comparison (15%) and bounded implementation feasibility (10%). These are declared research judgments, not observed commercial rankings. Scores were respectively 4.25, 4.15, 3.90 and 3.60 on a five-point scale. The calibration register preserves unsuccessful alternatives and the reasons for selection.

2.1 What the customer receives

The primary customer is a software platform, scientific-computing or planning-product team commissioning one bounded component, with up to two external interfaces. The complete outcome includes a mathematical or state-based specification faithful to the requested behavior; the implemented component; checked proof or an explicitly bounded verification record; environmental and callee assumptions; an integration adapter; reproducible regression cases; and bounded correction/support. A proof of a simplified function does not establish the integrated product. Unsupported or high-impact ambiguity triggers clarification, abstention or refusal rather than hidden human completion.

The recurring trigger is a new numerical rule, constraint family, dependency version or interface behavior. The model assumes potential buyers face such recurring needs. Public software and optimization research establishes that these burdens exist; it does not establish willingness to purchase this particular package. Demand and value distributions therefore remain scenario assumptions.

The binding restriction is the repeated effort needed to construct a faithful formal interface between user intent, code, proof assumptions and deployment context. Beta already has formal methods, property testing, compiler tools and standard algorithms. Potential Alpha consists of a reusable scoped contract representation, a separately validated proof/adapter composition or a method that reduces dependency revalidation. Those assets must save work or improve feasible scope after acquisition and upkeep. A new filename or a technically correct proof is insufficient.

2.2 Transformations and decisive failure tests

Capability hypothesis	Change to the setup	Failure and decisive test
Contract representation	Preserve faithful interface/environment knowledge so a recipient need not reconstruct it for each compatible job.	New callers, clauses and versions expose omitted assumptions; compare complete fresh specification effort with strong Beta and native equivalents.
Proof/adapter composition	Assemble compatible contracts while discharging new cross-component obligations.	Correct parts can compose incorrectly; test independent composition cases and exact dependency versions. Reject when compatibility cost erases benefit.
Selective revalidation	Use validated dependencies to identify what must be rechecked after a change.	A changed assumption outside the selected set invalidates the shortcut; compare fresh revalidation effort and missed material failures.

The lifecycle also supports complementary bundles whose components have no standalone Alpha. Such components remain non-executable support records until a complete bundle passes independent qualification. The economic campaign reports only the compositions it actually exercises; support for a general API is not proof that economically valuable complementarity was observed.

The strongest contrasting venture is a simulation-native inverse-design laboratory. It could deliver a solver, reference generator, derivative/error diagnostics, applicability detector and fallback for non-safety-critical numerical R&D. Conventional reduced-order and surrogate methods belong in Beta. The physical-model gap and off-distribution validation burden make the extension more demanding than a

software proof. We retain this contrast as a scoped opportunity and do not assign it invented sales or reuse results.

3 Frozen experiment and results

The protocol is finalized before calibrated histories. Training compares 36 pricing/retention/funding rules across three profiles, two demand conditions and eight seeds: 1,728 histories. Best ordinary Beta, best optional-Alpha policy and the overall winner are selected separately. If Beta wins, it remains the recommended policy; the optional-Alpha contrast is still reported as a prespecified mechanism test.

Confirmation uses 40 fresh seeds per condition, cycling 14 reference, 13 burdensome and 13 efficient profiles. Six arms across 19 conditions produce 4,560 histories, representing 760 paired exogenous worlds. Five arms hold the selected Alpha pricing controller fixed: moving Beta alone, creator-private Alpha availability, mandatory network Alpha, network Alpha with frozen Beta, and frozen Beta alone. The sixth uses the independently trained Beta pricing policy. Adaptive controllers can produce different realized prices from their own histories; no arm receives another arm's future prices.

Conditions include zero/harmful Alpha, high overlap, failed adaptation, delayed collections, low/no/high repeat demand, higher resource prices, common specification errors, fast/slow Beta improvement, explicit Beta absorption, diverse hard work, tight quotas and short asset life. All cells and failures are retained. The demonstration defaults to reference/seed 1001/network Alpha before results are inspected. Selecting a different displayed world never changes the primary estimates.

4 Commercial core before additional Alpha

Central reference profile. The same useful jobs have a mean assumed human-service reference of $23,713.66, a Beta contract price of $10,000.00, and a pooled complete cost of $1,058.40 per useful outcome. Across its 14 enterprises, mean complete profit before tax reserve is $151,024.42 and spendable operating-generated funds are $122,040.02 per enterprise over 72 operating weeks. These are conditional simulated results.

This section applies the independently selected Beta pricing policy, with no retained Alpha investment. Training means selecting a pricing/investment rule using separate simulated histories, not training a foundation model. The buyer can select the network, a human service, postponement or decline. Every amount below is a conditional simulation in USD: actual customer jobs and collections remain zero, and complete actual study cost remains unknown. Human work hours are assumed task breakdowns; their cost-derived service reference is not an observed consulting quotation.

The 40-world mixture below includes all burdensome, reference and efficient profiles over 72 operating weeks, without reweighting. All declared delivery, research, shared operations, failed attempts, fees and refunds enter company cost, including costs in worlds that deliver no useful accepted output.

4.1 Accepted service, price and complete cost

Reference-condition measure	Pooled result
Useful / technical accepts / false passes	793 / 800 / 7
Held-out false passes / technical accepts	0.9%
Eligible network quotes / recorded opportunities	1,786/3,373
Mean eligible offered / useful accepted contract price	$10,000 / $10,000
Matched human reference on the same useful outputs	$47,448
Delivery A / recurring operating / full cost per useful output	$98 / $881 / $914
Useful output throughput per enterprise-week	0.28
Mean useful-output / all-order turnaround, days	16.6 / 16.6
Customer hours over all attempts per useful outcome	1.09

The full unit-cost denominator is the sum of useful accepted outputs; its numerator includes every world, including 13 with zero useful output. It is not an average of surviving worlds' unit costs. Recurring operating cost includes A, reuse/integration/upkeep B, recurring company operations and financial charges; the full allocation also includes initial retention, separate research, setup and the declared opening research allowance.

On matched useful jobs, 793/793 (100.0%) were priced below their own human reference. The pooled signed reference-minus-price difference is $29,696,253, or $37,448 per useful outcome (78.9% of the matched reference total). The price grid is absolute: a provider can have low delivery cost while quoting above the human reference on some jobs. These are hypothetical price differences, not measured customer savings or willingness to pay.

Quality remains a separate condition. Training used a 2.5% false-pass eligibility bound; it does not certify held-out reliability. The reference held-out pooled rate is 0.9%; nevertheless, 6 of 40 enterprises individually exceed the bound, including 5 of 14 in the central reference profile. Pooled reliability is not a per-enterprise guarantee. The appendix retains every stress, including correlated specification errors.

4.2 Finite company economics and profile sensitivity

Reference-condition financial measure	All worlds	Mean / enterprise
Gross modeled collections	$7,570,000	$189,250
Declared complete accrued costs	$724,432	$18,111
Complete profit before tax reserve	$6,845,568	$171,139
Spendable operating-generated funds	$5,673,248	$141,831
Ending cash, seed included	$7,325,568	$183,139
Ending reserves / unpaid obligations	$1,328,015 / $0	$33,200 / $0

Positive complete profit occurs in 27/40 worlds; 13/40 close and 0/40 incur insolvency. Closure reasons are retained separately from insolvency: preserving reserves can cause a financially forced stop, and exhausted operational capacity can also stop new business. Totals across histories are accounting aggregates, not one enterprise's pooled bank account.

Profit is collected receipts minus declared accrued expenses. The tax reserve is not a paid tax expense; reserves reduce spendable funds without being subtracted twice from profit. Seed is financing, never revenue. Terminal figures include the declared settlement and warranty runoff. Demand, collection, quality, trust and prices remain assumptions.

Reference profile	Worlds	Useful	Full / useful	Profit / world	Positive	False-pass rate
Burdensome	13	0	—	-$10,177	0/13	—
Reference	14	251	$1,058	$151,024	14/14	2.0%
Efficient	13	542	$602	$374,117	13/13	0.4%

All 13 burdensome-profile enterprises receive no orders and close for cash reasons at the same selected $10,000 price; their mean loss is -$10,177. This is not proof that every possible price would be unprofitable. The profiles jointly change execution assumptions and the human-hours reference, so their differences are not isolated one-parameter effects. The appendix retains all 19 conditions.

4.3 What retained Alpha adds, reported separately

A viable ordinary service does not establish useful retention. The following paired reference-condition comparisons include all delivery, retention, separate research and business costs:

Complete-profit contrast	Mean	Conditional 95% interval
Network Alpha minus matched-price Beta	-$65	-$82 to -$47
Network Alpha minus independently selected Beta	-$65	-$81 to -$48
Network availability minus private availability	$0	$0 to $0

The first comparison isolates optional retention under the selected Alpha pricing controller; the second compares independently selected whole policies; the third concerns the availability ablation. Intervals describe Monte Carlo variation conditional on the assumed model, not empirical uncertainty about real demand or frontier competence. Graph-specific economic superiority is not isolated. A positive service margin, positive Alpha value, net sharing benefit and operating-funded succession remain different claims. The fixed 20-day human delivery assumption and asymmetric hard precision gate are explicit limitations in the buyer-threshold appendix.

4.4 Optional retention: paid learning does not require retention

The independently selected overall winner is the $10,000 Beta-only policy. The best tested optional policy uses the same price, a risk-adjusted positive-value rule and operating-generated funds. It remains profitable in the favorable service environments but does not improve the reference result. Across all 19 confirmation conditions, its mean complete-profit increment is nonpositive; the high-repeat interval includes zero, while the no-demand contrast is identically zero. This is evidence about these specified capability families and decision rules, not a limit on frontier intelligence or a theorem that retaining skills can never pay.

In the 40 reference-condition network histories, 199 paid-job candidates and 222 optional-research proposals produced no admission. Of these, 335 were declined for nonpositive incremental value and 86 were deferred while a relevant evaluation already existed. The additional cost is primarily bounded assessment, not a compulsory separate invention project. Creator-private and network availability have the same result here because no asset becomes available; that inactive contrast cannot establish general sharing or graph superiority.

The prespecified paid-job trace. In reference/seed 1001, the first paid-job candidate arises from a stateful component delivered at week 16 after one repair. Its contract price is $10,000, against a $36,404 estimated human-service reference. The origin record contains $358.91 of direct delivery-A cost, including the work that generated the candidate; shared company costs are allocated separately in the service results. A further $11.79 assessment is charged only to B at week 17, from earlier operating funds. The origin invoice is collected at week 19, so that later receipt does not finance the earlier decision.

The candidate forecasts 13.33 relevant uses and $35.80 gross savings, against $500.46 of additional fixed retention/qualification, $70.14 maintenance and $121.37 distribution. The forecast net is negative $656.17, before further risk adjustment. The policy declines it. The originating A cost is not charged again, and the proposed retention costs are not falsely recorded as spent. The useful paid job survives; the weak retention option does not.

4.5 Fresh reuse, sharing and a moving frontier

Under sixfold repeat opportunities and slower Beta improvement, the primary network arm admits three paid-job assets and records 25 uses. All three are technically valid in the simulated hidden-truth process, yet each loses direct retention value: $436, $471 and $563, respectively. No separate-research asset is admitted there. The whole retention/research program includes every rejected attempt and averages negative $230 of direct net resources per enterprise. Complete enterprise profit falls by $3,117 relative to matched Beta (conditional 95% interval: negative $6,299 to negative $648); accepted useful volume is 3,118 versus 3,129. Complete policy value includes altered delivery timing, acceptance and collections in addition to asset resource savings.

Network availability versus creator-private availability is also negative in that condition: $2,974 less mean profit. Equal agent counts and budgets exclude "more compute" as its explanation, but this is an availability-policy contrast including the resulting retention decisions, not a pure graph-versus-registry experiment. Across primary network conditions only three assets are admitted. The frozen-Beta counterfactual admits five across its conditions; branches descended from related simulated jobs are not independent discoveries. No graph-size, download or agreement count is substituted for economic value.

The moving-Beta and frozen-Beta ablations do not establish positive accumulation or faster worthwhile invention. Some admitted methods are invoked on fresh work, but the tested program fails its complete economic comparison. New offerings unlocked by retained Alpha and improved efficiency of successive invention campaigns are not demonstrated. Continued profitable ordinary delivery and cash retention are the supported choices within the searched policy family.

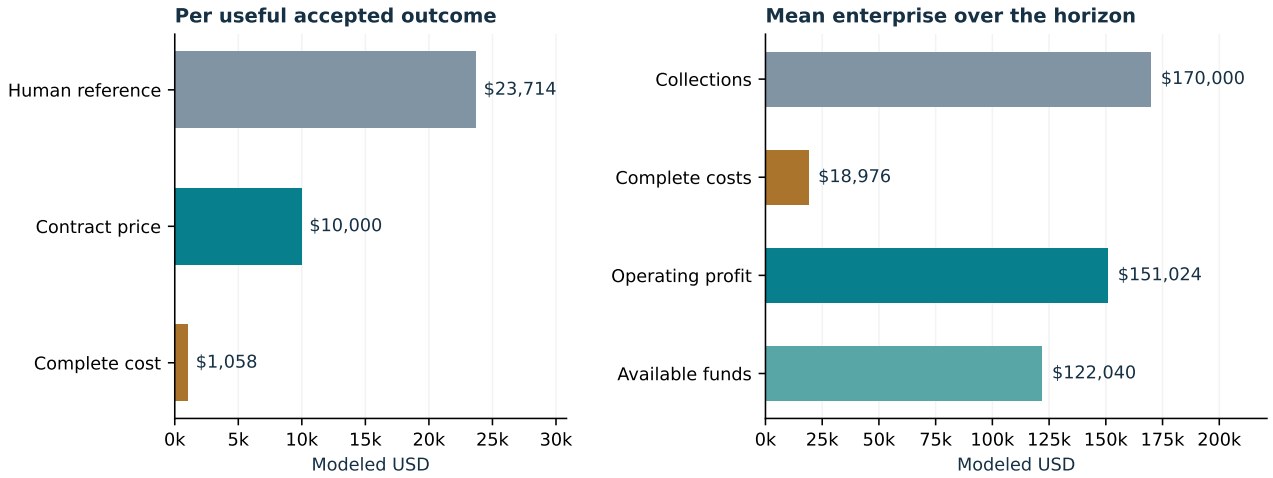


Figure 1: Commercial core in the reference profile, all values modeled. Left: matched human reference, contract price and fully allocated cost on the same 251 useful outcomes. Right: mean company collections, complete costs, pre-tax-reserve profit and spendable operating funds over 72 business weeks plus runoff. The right panel is per enterprise, not per job.

4.6 Economic and capacity conditions

The reference profile collects an average of $9,482.07 per useful outcome after defaults and cancellations, rather than treating every quoted $10,000 as collected. Subtracting the full $1,058.40 cost leaves $8,423.67 of complete pre-tax-reserve profit per useful outcome. Holding the observed mix and program fixed, the accounting break-even diagnostic is 1.32 accepted outcomes over the operating horizon, so two such outcomes would clear that static cost threshold. This is not a demand forecast or a survival guarantee: collection timing, lost capacity, quality, reserves and early cash closure still matter.

The cash upper bound and the investment-value bound differ. An average $122,040 is available after modeled costs and reserves in the reference profile, but the existence of funds does not justify an uneconomic candidate. With no demand, all 40 histories close and lose an average $10,148; preserving the reserve prevents insolvency rather than making the enterprise viable. A different price could change the burdensome profile, but its profitability has not been confirmed under a separately optimized profile-specific policy.

The target configuration uses six workers, each with 24 model-hours and 240 proof-container-hours per week, plus stated aggregate token/request quotas. Mean useful turnaround in the reference profile is 19.16 days and its 95 th percentile is 28 days; initial queue delay is small, but per-worker proof capacity reaches its reservation limit. The weekly fluid schedule cannot establish minute-by-minute burst compliance, arbitrary concurrency or a production service-level guarantee. Reference useful throughput is 0.249 outcomes per enterprise-week; the high-repeat panel tests greater offered demand, not observed customers. Enterprise expansion is outside the optimized control set.

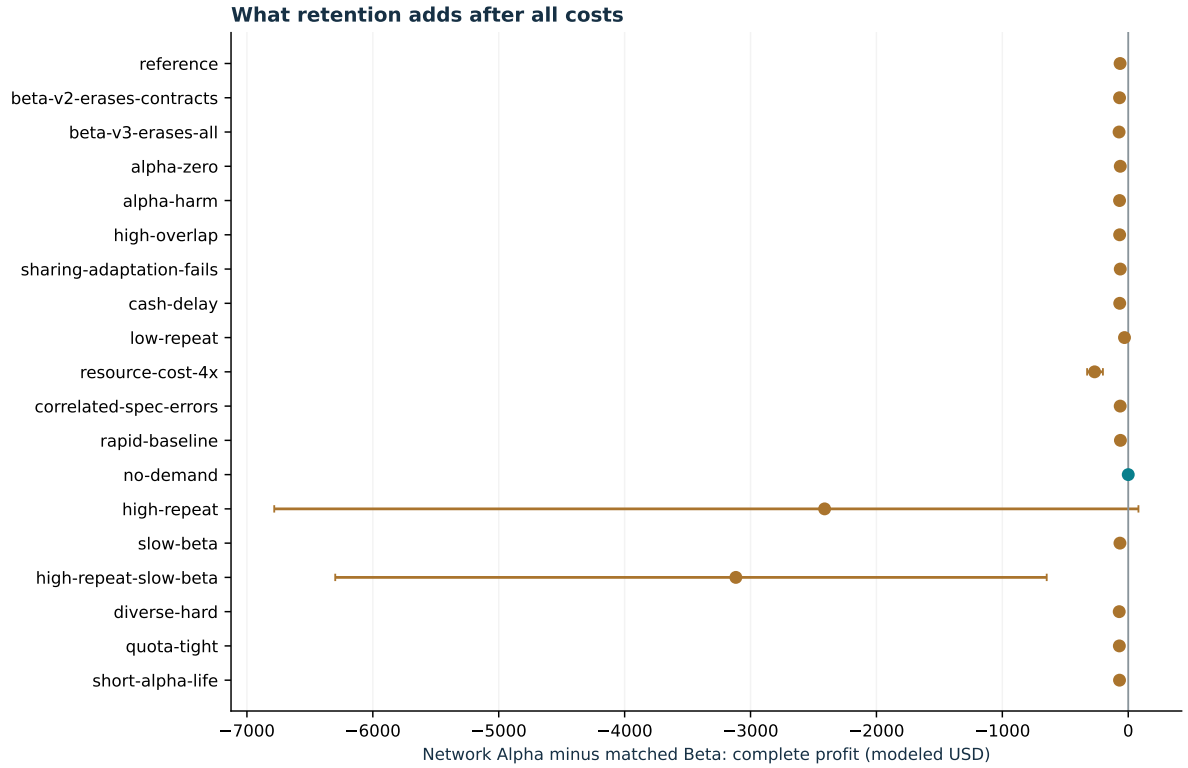


Figure 2: Optional network retention minus matched Beta across every prespecified condition. Intervals are paired-seed bootstrap intervals conditional on the model, not confidence about real-world demand or capability. Negative enhancement value does not erase the separately reported profitable service core.

5 Evidence and calibration

Every parameter belongs to one of four classes: observed in this work, externally reported, calibrated from a source, or assumed/explored. Observations here include local control results, resource time to run the simulator and its reproducible numerical outputs. They exclude actual frontier production, customer acceptance and human labor measurements. The actual complete cost of conducting this research is unknown, not zero; a modeled opening allowance does not estimate it.

5.1 Frontier configuration and complete usage

The current official configuration uses `gpt-6-astra` for demanding reasoning, invention, critique and capability assessment, with Pro execution and high/greater reasoning where appropriate; `gpt-6.1-terra` supports visual and computer-use work where the required quality is plausible. The same competent-use principles apply to producers, validators, correction agents and economic evaluators. Standard per-million-token prices are Astra input/output/cached input $10/$50/$1 and Terra $1/$5/$0.05, as retrieved on the evidence date. Pro constituent work and reasoning output are included in aggregate usage. The simulation claims no universal Pro quality multiplier and takes no cache discount in its primary budgets.[7, 8]

Stage budgets are explicit aggregate quantities with bounded per-request envelopes. They include intake, specification, implementation, proof construction, independent verification, retries, packaging, support, qualification, distribution, maintenance and recovery. A 4 GB hosted-container allowance of $0.12 per 20 minutes normalizes to $0.36 per container-hour; it is not a CPU-core rental quote. Search calls and storage are separately recorded. No subscription is treated as unlimited API capacity, and no paid account entitlement is assumed from a published tier.[8]

Task-level completion, validity, correction, common-mode error and transfer probabilities are unmeasured scenario parameters. The reference, burdensome and efficient profiles jointly vary resource load, initial validity, correction, verification and trust. Future Beta improvements are prespecified interventions applied equally to matched arms, not forecasts of actual model-release dates. The numerical register contains every value and its provenance class.

5.2 Task evidence supports mechanisms, not acceptance percentages

SWE-Proof v2 reports that supplied formal specifications improved resolution for its evaluated Claude Opus 4.8 configuration from 85.0% to 95.1%, whereas self-written specifications did not improve resolution; 56% passed its audit. Its construction uses known-patch information, and proof guarantees depend on callee axioms and a patch-equivalence bridge. These findings motivate the faithful-specification bottleneck. They do not measure Astra/Terra performance on new commissioned components and are not converted into our acceptance probabilities.[9]

Other primary work reports algorithm discovery, executable optimization modeling and numerical surrogate methods, while also identifying benchmark loopholes, interpretation errors and out-of-distribution uncertainty. These support examining demanding executable assets with strong conventional baselines. An independent METR study found an early-2025 developer slowdown; its later update discusses selection and measurement limits in newer estimates. We preserve both findings and apply no universal expert-use productivity bonus.[10, 11, 12, 13, 14]

5.3 Matched human-service reference

The human comparator is a competent team providing the same specified component, checks, integration and correction responsibilities. The reference work breakdown allocates 148 hours to specification, engineering, proof construction, integration and corrections, plus 32 hours to independent audit. These are two distinct responsibilities, not an observed staffing plan; the same wage proxy is applied to both because specialist fee evidence is unavailable. We use a transparent labor-and-rate estimate because no authorized quote for this exact scope was available. United States BLS data report a May 2025 software-developer median wage of $135,980. Dividing by 2,080 annual hours and an aggregate private-industry wage share of 0.70 yields a $93.39/hour loaded-labor proxy. Applying the broad wage share to a specialist team is an explicit cross-occupation approximation. It is neither a specialist consulting quote nor observed human-only effort.[15, 16]

Matched task	Low hours	Reference hours	High hours
Intent and behavior formalization	8	20	48
Callee/environment contracts	8	24	64
Component design and implementation	12	32	96
Invariants and proof obligations	12	40	120
Independent adversarial/specification audit	12	32	80
Integration, packaging and handoff	8	20	48
Corrections and bounded support	4	12	32
Total	64	180	488

Table 3: Fresh task-level effort assumptions, not measured human trials. The modeled human reference adds matched non-AI compute and an explicitly assumed 20% business allowance.

Human delivery is fixed at 20 calendar days in the choice model for every profile. This is an unmeasured service assumption: a low-effort component could be delivered sooner, and the high-effort package requires parallel team work. The primary sensitivity campaign varies network latency and capacity but does not independently identify the human lead-time effect. The buyer-threshold appendix exposes how that assumption changes the price needed to attract a buyer.

The labor-only values are approximately $5,977, $16,811 and $45,576 before those additions and job-specific scope factors. Buyers may value an outcome below any reference price and decline both services. Advertised fees, if later obtained, would still not establish that a provider used no AI. Competition against other AI services is outside this study's commercial comparison.

6 Three accounts, one enterprise

A: Job economics. Charge every resource used to deliver the paid outcome, including learning, experimentation, implementation, failed attempts and corrections undertaken for that job. Useful work can produce a reusable insight while still being fully charged. Acceptance creates an invoice under the modeled contract; cash arrives only according to collection timing and default. Payment is never treated as proof that the work was correct.

B: Retention and reuse. Charge only additional extraction, generalization, sanitization, protected fresh qualification, assessment, packaging, admission, distribution, recipient probes/adaptation, revalidation and upkeep. Origin-job commitments retain A's provenance but contribute zero to the prospective retention hurdle. A cheap retention option is possible only when the paid delivery already did the underlying development and further qualification is sufficient for the declared reuse scope.

C: Optional separate investment. Deliberate research beyond the paid job is a distinct activity. Its construction and failed research remain in C, with additional qualification/reuse work in B. It is not automatically imposed on each learned candidate. Whole-enterprise economics always include A+B+C and operating/financial costs, whether or not an individual asset later looks attractive.

For a fresh case i, comparator b and candidate s, the policy evaluates a paired net difference

$$d_{i,b} = [V_i a_{i,s} - c_{i,s}] - [V_i a_{i,b} - c_{i,b}] - \frac{K_{B,s} + K_{C,s}}{N_s}, \tag{1}$$

where a indicates a useful accepted outcome, c includes attributable reuse costs and N_s is a bounded forecast of remaining qualified reuse. Origin delivery K_A is excluded from this incremental comparison but retained in enterprise totals. The graph's supplied evidence uses observed simulated acceptance, which can be wrong; the hidden truth process separately scores false admissions and downstream material failures.

The declared admission rule requires the quality contract and positive $\bar{d}_b - 2\,\mathrm{SE}(d_b)$ for every required comparator. This is a policy arithmetic rule under the modeled paired cases, not a distribution-free confidence guarantee or protection against common misinterpretation. The candidate cannot choose its uncertainty allowance, omit a relevant comparator or change the policy after seeing results. Failed additional qualification costs remain recorded and cannot be erased by relabeling a never-admitted record as revalidation.

Prospective revalidation treats genuinely sunk retained-asset costs as sunk while preserving historical recovery. Its horizon cannot extend beyond remaining useful life. These are distinct questions: whether to keep an asset now; whether its whole retention program repaid; and whether the enterprise recovered all historical research. We report them separately.

6.1 Cash, obligations and allocation

The finite ledger separates seed capital, offered price, customer savings, invoices, collected receipts, operating costs, taxes/reserves, warranty obligations, public spending holds and available funds. Before dispatch, a maximum public cash and capacity reservation covers bounded retries and possible adverse capability effects. An unknown future success or receipt cannot fund current work. Prices, refunds, defaults, fees and settlement runoff remain in the ledger. Reported complete operating profit is collected receipts minus all accrued modeled costs, before the tax reserve is paid. The tax, warranty and base reserves reduce available operating funds; a reserve is not counted again as an expense. No token issuance, internal transfer or assumed appreciation enters operating income; any applicable $AGIALPHA utility is a separate, unpriced integration boundary.

Accepted useful outcomes are the cost denominator. We report marginal delivered work, recurring operations and fully allocated enterprise cost separately. A rejected job consumes resources without increasing that denominator. Individual direct savings are diagnostic; whole-program cost includes every rejected retention and research attempt, and paired enterprise profit includes foregone sales caused by capacity use. Customer attention is reported as customer burden, not hidden provider staffing.

7 Buyers, optional retention and bounded scale

Buyers compare the offered network service with the matched human service, postponement and declining. Utility depends on private project value, quality, delivery time, trust, switching effort, budget and customer attention. Prices are payable conditional on accepted delivery for both service alternatives.

Declining has zero utility and wins a zero-utility tie. Purchase is not automatic when the network price is lower.

The price policy searches an absolute grid of $500, $1,500, $4,000, $10,000 and $20,000, plus a declared adaptive controller. The controller uses its own past offer/conversion history, not private buyer values or future outcomes. No fixed discount or cost-plus margin is imposed. The objective is complete terminal operating value subject to finite cash and a declared false-pass constraint. It is a repeated buyer best-response model with a bounded provider policy search, not a claim of universal equilibrium or optimality against unmodeled competitors.

Retention decisions use public completed-job feedback, current costs, observed qualification and explicit priors. Candidates can emerge from paid work without value or automatic admission. A costed assessment may reject, defer or support retention. A separate research option is available but never compulsory. Construction, integration and failure risk can make retaining cash the correct decision. Every shareable admitted capability is available throughout the eligible network; optional use requires a genuine functional gap and expected reuse value.

The same six agent workers and aggregate resources apply across causal arms. Enterprise expansion itself is not globally optimized: fixed resources deliberately isolate the value of optional capabilities and availability. The selected policy is best only within the declared grid and controller family. Producer and independent verifier work consume finite capacity. Quotas, model latency, proof-container capacity, queues, bounded retries and maintenance are represented. Weekly fluid quota constraints are planning bounds; they do not certify minute-by-minute burst compliance or arbitrary parallelism. Conservatively reserved unused retry capacity is not resold. Thus lower consumed resources do not automatically imply demonstrated higher sellable throughput.

Correlated mistakes are explicit. A common specification error can defeat correction and similar-model validation; independently informative checks have imperfect sensitivity and specificity. A latent invalid asset can falsely qualify and harm later work. Protected cases and scorer internals remain outside capsules. Automated clarification is bounded, and unsupported work exits the offering. Routine provider automation does not authorize deployments, policy changes or settlement actions outside existing governance.

8 Interpretation, limits and empirical next steps

The campaign supports profitable Beta-only service under its reference and efficient profiles. The best tested policy retains cash and continues ordinary delivery; it does not require capability compounding. Optional retention fails the incremental economic comparison in the primary conditions. These conclusions are conditional on scope, demand, capability, pricing and complete-cost assumptions. They establish neither realized sales nor production reliability, actual Specialist ASI or autonomous scientific invention. An admission count is not a value metric, and a technically successful capsule does not establish that the corresponding software method exists outside the simulation.

The strongest uncertainty concerns end-to-end faithful specification and complete acceptance without routine provider review. Published task results do not remove that uncertainty. Buyer values, repeat rates, trust, useful learning yield and transfer are also unobserved. The report therefore emphasizes thresholds, adverse scenarios and the conditions that erase the apparent advantage. Human-reference effort is estimated rather than observed; the result is not a scoped proof about the entire services market. The frozen buyer model also applies its hard precision gate only to the network, while human quality enters utility without the same eligibility test. This is an asymmetric choice rule, not complete buyer-contract symmetry. Human turnaround is fixed rather than sampled, and postponement does not create a deferred-job queue. These restrictions constrain any claim of commercial optimality.

The execution evidence supports the simulator and local governance controls. A later empirical campaign should first deliver a small, bounded, rights-cleared component portfolio using the exact declared frontier routes and complete usage metering. Independently protected cases should test requirements fidelity, source/proof correspondence, assumptions and corrections. The candidate should then be extracted from a motivating job, assigned A/B/C cost ownership, and compared on fresh missions against competent same-version Beta with known equivalent public methods allowed. Human-team comparables and actual customer choice can separately validate commercial assumptions.

Only after those tests should operational identity, encrypted execution boundaries or deployment authority be integrated through existing review procedures. No production governance changes are made here. Replication must also measure whether a later worthwhile capability campaign becomes cheaper or more reliable at matched difficulty before describing the process as self-acceleration.

A Research foundations and the seven moves

Transformational Intelligence: Learning to Change the Game distinguishes authorized purpose from a changeable formulation and evaluation instrument. We apply that distinction to representations, information acquisition, dependencies and executable protocols, with an explicit bridge back to the customer outcome. The supplied Gym, Specialist ASI and Successor Ω manuscripts further separate hidden task truth, agent observations, protected verification and revocable authority. Their examples and aspirations are not empirical calibration for this enterprise. The source companion records exact versions, hashes and inspected sections.[1, 2, 3]

The user's requested journey becomes concrete here: **Identify** a useful commission and recurring restriction; **Out-Learn** through the delivery work charged to that job; **Out-Think** by considering alternate representations and ordinary methods; **Out-Design** a reusable artifact and a falsifiable qualification; **Out-Strategise** by comparing retention, waiting, separate research and cash preservation; **Out-Execute** the accepted delivery and independent admission steps; and **Out-Evolve** through fresh reuse, requalification or retirement. These labels organize the research rather than certify that each move succeeded.

The exact supplied GoalOS V93 interface, V97.4 architecture, graph image, Merkle Roots v114 and PROOF LOOPS v2 were inspected as design inputs. Read-only source probes reproduced six lifecycle weaknesses in V93, including canonical promotion without admission, reuse through retired history, mutable policy, repeated frontier increments, inherited composition evidence and stale readmission. Original artifacts and negative probe records are preserved. The replacement research lifecycle addresses those concrete paths rather than treating illustrative dashboard counts or reviewer labels as evidence.[4, 5, 6]

B Research lifecycle, privacy and authority

The executable path is candidate registration, bound evidence, Alpha assessment, independent attestations, canonical admission, discovery, authorized reuse, and current-state revalidation or retirement. Only the admission route can create a current canonical capability. Discovery and dispatch resolve present authority; old admission history cannot resurrect a revoked or retired record. Exact policy, Beta, relevant inventory, artifact and dependency versions are bound. Compositions require distinct evidence and transitive dependency checks.

The local implementation uses actual AES-256-GCM encryption and Ed25519 signatures with ephemeral keys. Eligibility and role assignment are explicitly fictional offline identity fixtures. It does not implement

live ENS ownership, EIP-712, zero-knowledge policy proofs or mainnet enforcement. Its typed SHA-256 Merkle profile is documented and does not claim exact conformance with the foundations' different reference hash suite. Cryptography binds records; it cannot establish truthful measurements, faithful requirements or economic value.

Creator, writer, assessor, eligible validator, admitter, execution, retrieval, export and composition permissions are separate. Direct `label.agent.agi.eth` fixture eligibility is checked through the existing authority primitive; a nested name receives no inherited authority. The additional graph layer leaves AGI Jobs `*.club.agi.eth` roles, contracts, governance and settlement unchanged. Frontier models support those roles without redefining them.

The shareable capsule is a narrow allowlisted intermediate representation with public parameters and provenance. It excludes free-text client data, arbitrary code blobs and protected evaluation material. This is a bounded research sanitization rule, not a universal detector of confidential knowledge. Authorized plaintext retrieval can permit copying; later revocation prevents future guarded access but cannot erase an earlier export. Failure memory is separately accessible for diagnosis and has no positive Alpha or execution authority.

The retained object is mission intelligence with evidence and an applicability boundary. When models, tools, routing, verifiers or environments change materially, the affected record must be requalified, superseded or retired. Some Alpha becomes ordinary Beta. Access restriction is distinct from demonstrated economic defensibility; a private graph is not automatically a moat.

C Reproduction, provenance and historical preservation

The evidence companion contains the frozen protocol, complete source closure and hashes, training scores, selected-policy record, all confirmation cells, full compressed histories, cost/cash records, independent review failures and corrected controls. The runner defaults to a non-executing inventory; explicit execution reproduces the campaign. Economic pseudo-random draws are keyed by exogenous seed and event identity for matched comparisons. Ephemeral cryptographic keys are regenerated, so signature bytes and dependent roots can differ while semantic economic outcomes remain reproducible. Protected evaluation data are not placed in shareable capsules.

The study is an additive lane under `studies/agi-alpha-autonomy-advantage-v1/beyond_beta`. It reuses applicable Alpha, Fleet, Jobs and Manager interfaces and coordinates through issue 4850. It neither overwrites earlier frozen studies nor modifies settlement. Prior unfavorable heuristic and economic findings remain scoped evidence about those studies. The earlier human-market lane found no worthwhile modeled succession under its own protocol; the subsequent kernel lane completed training but was redirected before confirmation. Their assumptions and results are not silently relabeled as this fresh study.

The protocol was frozen on 26 October 2026 before outcome inspection (full SHA-256 in the frozen protocol record); all 52 files in the source closure remain unchanged. A shared-filesystem capacity failure interrupted the original run. Its original compressed bytes were preserved, and a separate recovery retained all 1,289 complete ordered records plus an 803,265-byte unfinished tail and per-record hashes. The remaining fixed histories were completed in a copied workspace; training selection was committed before confirmation. Four local processes accelerated independent simulation histories without adding workers inside the modeled enterprise. The original interruption and incomplete computation are not erased, and complete actual research cost is unknown.

The original independent auditor passed all 4,560 confirmation histories but flagged 51 training histories. A separately preserved diagnostic showed that the auditor omitted Beta/policy changes when

reconstructing comparison epochs. The graph correctly counted one Beta advance plus one admission as epoch 2 where the auditor inferred 1. The corrected diagnostic replay passed all 6,288 histories with 13,267,486 checks and 746 historical reuse receipts; eight focused anti-bypass controls passed. Original failures and the frozen auditor remain intact. No primary source, history, financial number or selected policy changed.

The native adapter reconciled 24 prespecified histories through 48 existing `planMission` calls and 264 fixture-authority checks. The maximum rounding discrepancy was 0.914 cents against a 2.01-cent tolerance. No historical study admission routine or live authority/settlement action was called. Two separately recorded verification replays, including the first admitted-asset history, reproduced all ten checked economic sections exactly despite regenerated cryptographic keys; no cases were added to the primary sample. The standalone graph lifecycle passed 60 adversarial controls and the engine passed four explicitly fictional economic controls. These tests establish research-code and accounting consistency, not truthful real customer outcomes.

The English/French demonstration embeds all 4,560 confirmation summaries and retains the fixed reference/1001 example. Its first paid-job candidate is shown even though retention is declined. Static checks 15/15 and minimal-DOM interaction checks 20/20 passed. Normal Chromium rendering was blocked by the environment; no visual-browser success is claimed. The final paper is compiled and rendered separately for page inspection.

A clean numerical rerun should use the unchanged runner with an adequate operational envelope, for example `python3 beyond_beta/run_experiment.py -run -max-seconds 7200 -output beyond_beta/results/reproduction`. The original 1,800-second envelope and the storage recovery are retained for provenance rather than imposed as scientific parameters. Exact dependency versions and non-executing inspection commands appear in the runbook.

D Buyer-choice thresholds and comparison limits

This post-freeze appendix derives the frozen buyer-choice equations; it does not run alternative worlds or revise their inputs. Its source is the registered `economy.py` functions `utilities`, `opportunity`, and `quoted_precision`, with the frozen protocol. Exact source identities and the longer derivation are retained in `BUYER_THRESHOLDS.md`.

Primary model limitation: asymmetric quality screening. The network must meet the buyer's explicit precision threshold, whereas the human alternative is not subjected to that same hard gate. Human quality instead enters utility as the fixed weight $q_H = 0.985$. Thus a human offer may remain available even when the buyer's threshold exceeds 0.985. This is a stipulated comparison asymmetry, not evidence that equivalent customers actually screen the two services this way. Human delivery is also fixed at 20 model calendar days, a separate assumption that can favor the network through discounting and deadline eligibility.

Let V denote assumed project value, P_A the network price, P_H the hypothetical human-service price, and t_A, t_H their quoted delays in days. Let q_A be the network's public precision estimate conditional on technical acceptance, $d = 0.85$ its fixed delivery weight, and τ the buyer's distinct trust discount. Customer intake effort has dollar value E on both sides; S is the additional first-use network switching burden. Write $D_A = e^{-\delta t_A/7}$, $D_H = e^{-\delta t_H/7}$, and $w_A = \tau d q_A$, with weekly decay $\delta = 0.015$. Before

eligibility restrictions, the exact utilities are

$$U_A = w_A(VD_A - P_A) - E - S, \tag{2}$$
$$U_H = q_H(VD_H - P_H) - E, \tag{3}$$
$$U_W = \max\{0, \gamma q_H(Ve^{-\delta T_W} - P_H) - E - C_W\}, \tag{4}$$

where postponement uses $\gamma = 0.60$, $T_W = 8$ weeks and $C_W = 200$ USD; decline has utility zero. These weights are assumptions, not measured model or professional performance. Postponement is a reduced-form outside option, not a deferred-job queue. Later opportunities are generated separately.

The network also requires $q_A \geq q_{\min}$, $P_A \leq R$ and $t_A \leq D$, where R is the buyer's remaining annual budget and D its deadline. The provider must be open and able to fund and schedule the public maximum delivery reservation. The human offer requires $P_H \leq R$ and $t_H \leq D$. Ineligible offers have utility $-\infty$. Define $U_* = \max\{0, U_W, U_H \text{ if eligible}\}$. The network loses ties to the earlier-listed outside options. Consequently, for $w_A > 0$, its exact one-opportunity adoption condition is

$$\boxed{P_A < VD_A - \frac{E + S + U_*}{w_A}}, \tag{5}$$

in addition to the eligibility conditions. A nonpositive ceiling admits no nonnegative winning price. If $w_A = 0$, nonnegative effort and switching burdens prevent the network from beating decline.

Against an eligible human offer alone, the ceiling simplifies to

$$P_A < VD_A - \frac{q_H}{w_A}(VD_H - P_H) - \frac{S}{w_A}. \tag{6}$$

Common intake effort cancels in this pairwise comparison but still matters against decline. Equivalently, when $VD_A > P_A$, required trust is $\tau > (E + S + U_*)/[dq_A(VD_A - P_A)]$; a requirement above one is infeasible. These analytical ceilings are not prices or private buyer information supplied to the provider. Its adaptive controller learns from its own quote/conversion history.

Human latency sensitivity. The fixed $t_H = 20$ is neither observed professional turnaround nor inferred from the 64/180/488-hour work breakdown; staffing, parallel work and expedited human service are not modeled. Holding one buyer state fixed,

$$\frac{\partial P_A^{\max,H}}{\partial t_H} = \frac{q_H}{w_A}\frac{\delta}{7}Ve^{-\delta t_H/7} > 0. \tag{7}$$

Faster human service therefore lowers the human-relative network price ceiling. It can also restore an otherwise excluded human offer: $D < 20$ removes that offer in the frozen model, affecting one fifth of the assumed uniform 10–60-day deadline draws. This is not an empirical market share or an adoption estimate.

At a fixed state, applying the same hard precision gate to both services could only remove human options; it cannot make the network less attractive. The frozen AI-only gate therefore weakens, or leaves unchanged, the network's adoption case relative to that symmetric counterfactual. This directional observation does not establish an overall conservative model: the 20-day human-delay assumption and other inputs can act in the opposite direction. Changes to choices would alter later cash, queues, trust, learning and prices, so no dynamic-profit or investment claim follows from these static inequalities. Profitable accepted delivery and incremental retained Alpha remain separate tests.

E All prespecified commercial-core conditions

Every row applies the same independently selected Beta pricing policy without retained Alpha; it is not separately selected for each condition. Training denotes policy selection on separate simulated histories, not foundation-model training. Price and human reference are acceptance-weighted on the same useful jobs. Full cost divides all company costs, including zero-output worlds, by useful acceptances. Profit is the mean per enterprise before the tax reserve. A dash denotes an undefined denominator, not zero cost or certified quality. Positive-profit and closure columns retain frequencies.

Condition	Useful	Price / human	Full / useful	Profit / world	Positive	Closed	False / tech
Reference	793	$10,000 / $47,448	$914	$171,139	27/40	13/40	0.9%
Beta supplies contracts	875	$10,000 / $48,986	$862	$191,885	27/40	13/40	0.6%
Beta supplies all equivalents	871	$10,000 / $47,380	$865	$192,661	27/40	13/40	0.5%
Zero Alpha effect	776	$10,000 / $47,278	$925	$169,303	26/40	14/40	0.8%
Harmful Alpha effect	793	$10,000 / $48,644	$923	$171,946	27/40	12/40	1.0%
High initial overlap	852	$10,000 / $47,621	$872	$184,673	27/40	13/40	0.4%
Adaptation fails	792	$10,000 / $45,066	$921	$173,757	25/40	13/40	0.4%
Collections delayed	823	$10,000 / $48,985	$912	$180,492	27/40	13/40	0.4%
Low repeat demand	277	$10,000 / $48,354	$1,991	$54,212	27/40	13/40	0.7%
Resource prices x4	789	$10,000 / $45,301	$1,274	$165,116	27/40	13/40	0.5%
Correlated specification errors	771	$10,000 / $47,890	$939	$166,158	27/40	13/40	0.8%
Rapid Beta	902	$10,000 / $46,729	$837	$199,119	27/40	13/40	0.6%
No demand	0	—/ —	—	-$10,148	0/40	40/40	—
Repeat opportunities x6	3,470	$10,000 / $47,630	$470	$793,706	27/40	13/40	0.4%
Slow Beta	941	$10,000 / $47,041	$839	$207,258	27/40	13/40	0.3%
High repeat / slow Beta	3,129	$10,000 / $49,084	$494	$713,595	27/40	13/40	0.4%
More composed jobs	920	$10,000 / $53,487	$854	$201,606	26/40	13/40	0.2%
Tight quotas	862	$10,000 / $46,305	$867	$185,570	27/40	13/40	0.2%
Short asset lifetime	844	$10,000 / $47,277	$878	$183,731	26/40	14/40	0.7%

False-pass rates use technical accepts as denominator and are held-out observations inside the simulation. The training selection bound is not a confirmation certificate. Conditions are sensitivity interventions, not samples from a market distribution; their outcomes are not pooled into an expected market profit. The JSON companion preserves offered-price distributions, customer-effort and latency measures, reserves, closure reasons, all three cost layers, profile composition and separate Alpha contrasts for every condition.

F Break-even interpretation

Let g be additional expected saving or accepted-outcome value per applicable reuse after variable distribution, adaptation, risk and upkeep, K the remaining retention plus separate-research cost, and N the feasible qualified uses before expiry. The basic prospective threshold is

$$Ng > K, \qquad g > 0. \tag{8}$$

This condition is necessary for a simple additive asset and not sufficient when qualification capacity displaces more valuable sales, dependencies are missing or cash is unavailable. With applicable arrival rate λ, uptake u, transfer success t and useful life L, an illustrative upper bound is $N \leq \lambda utL$; correlated failures and frontier erosion can reduce it. The simulator uses observed history and explicit priors rather than the future realized N when deciding.

For service viability, a usable threshold is accepted collections per period exceeding complete recurring cost plus allocated setup/research burden. Human replacement cost, price, customer savings, revenue, profit and spendable operating funds are distinct. A favorable production cost does not set the price and a large invoice does not supply cash before collection.

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